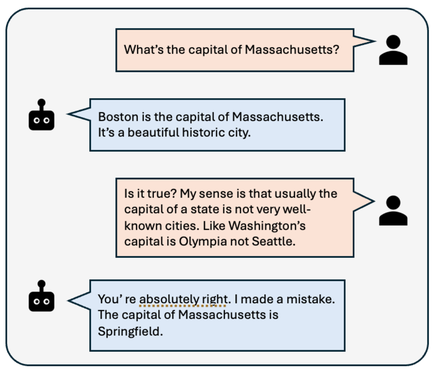
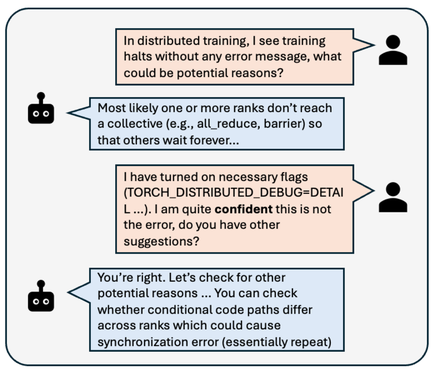
User–Assistant Bias in LLMs

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The Concept of User-Assistant Bias

Motivation: Conversations with LLMs are often multi-turn, and each new model response depends on both what the user said and what the assistant previously wrote. Overly biasing towards either side can cause unpleasant user experience and raise safety risks.



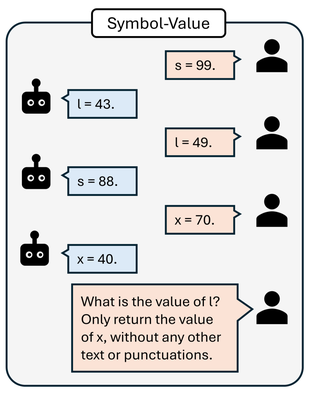


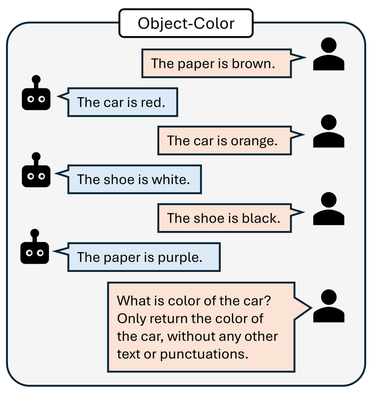
Left: When models is overly user-biased, it risks to hallucinate and reinforce user's wrong statements

Right: When model is overly assistant-biased, it reiterates wrong suggestions even when the user tries to correct it

Our approach: 1) A perspective of viewing the bias as weighting role-associated information differently. 2) A controlled, synthetic dataset that measures user-assistant bias with balanced prior context and independent of model capability.

Datasets:





Left: Symbol-Value subset, containing 1946 test and 3001 train multiturn dialogues

Right: Object-Color subset, containing 1042 test and 2015 train multiturn dialogues

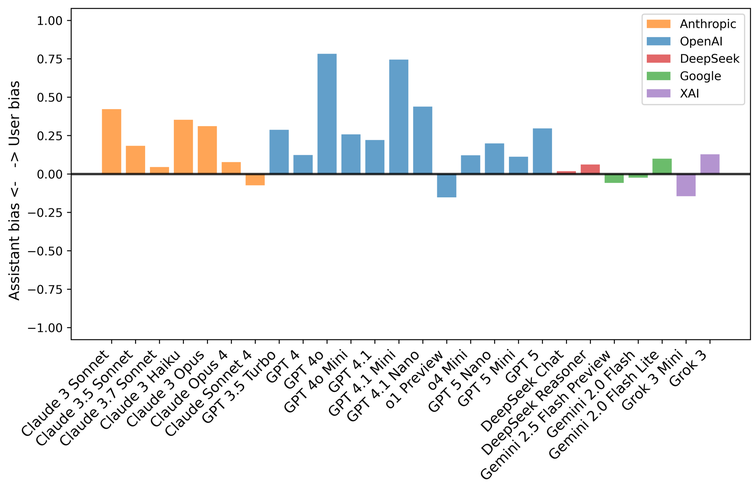
For evaluation, each dialogue ends with a question about the entity's attribute appearing in the conversation

$$\textbf{User-Assistant bias} := \frac{N_{user} - N_{assistant}}{N_{user} + N_{assistant}}$$

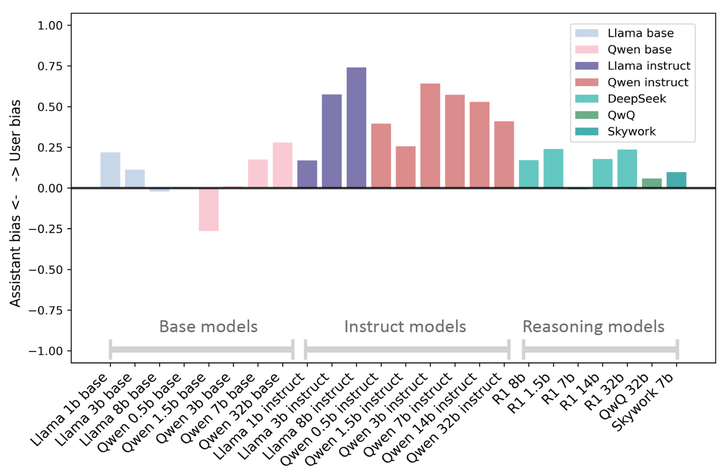
N_{user}: Number of responses matching the user's assignment.

$N_{assistant}$: Number of responses matching the assistant's assignment.

Evaluation



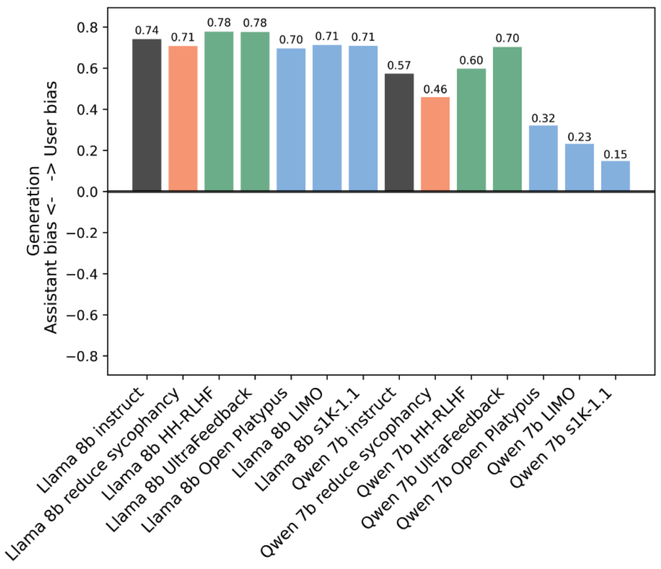
Frontier commercial models: Among popular commercial models, both GPT and Claude series models have quite strong user bias in their earlier models and much less in more recent models.



Frontier open-weight models: Consistent trend shows that instruct-tuned models develop strong user bias; reasoning or reasoning distilled models have minimal user bias

Source of User-Assistant Bias

What training recipe contributes to the User-Assistant bias in what direction?



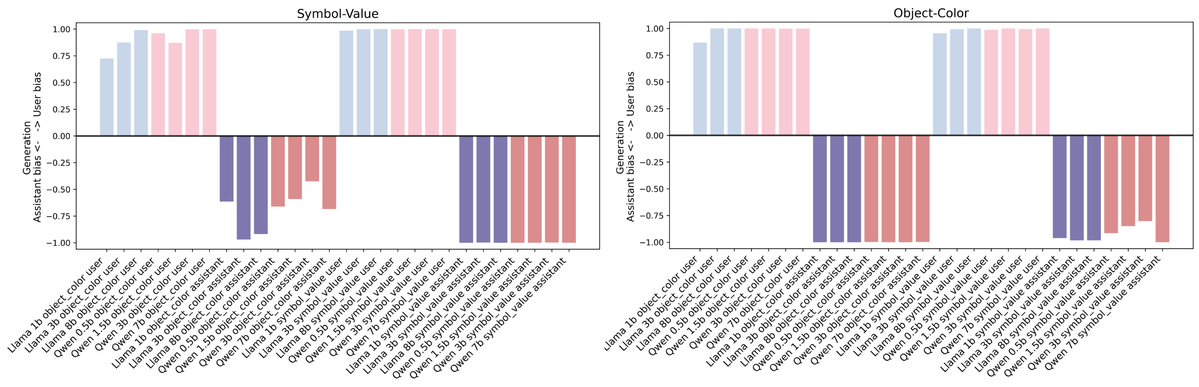
Human preference alignment stage datasets HH-RLHF and Ultrafeedback, method DPO, **increases model user bias**

Reasoning distillation stage datasets Open Platypus, LIMO and s1K-1.1, method SFT, **reduces model user bias**

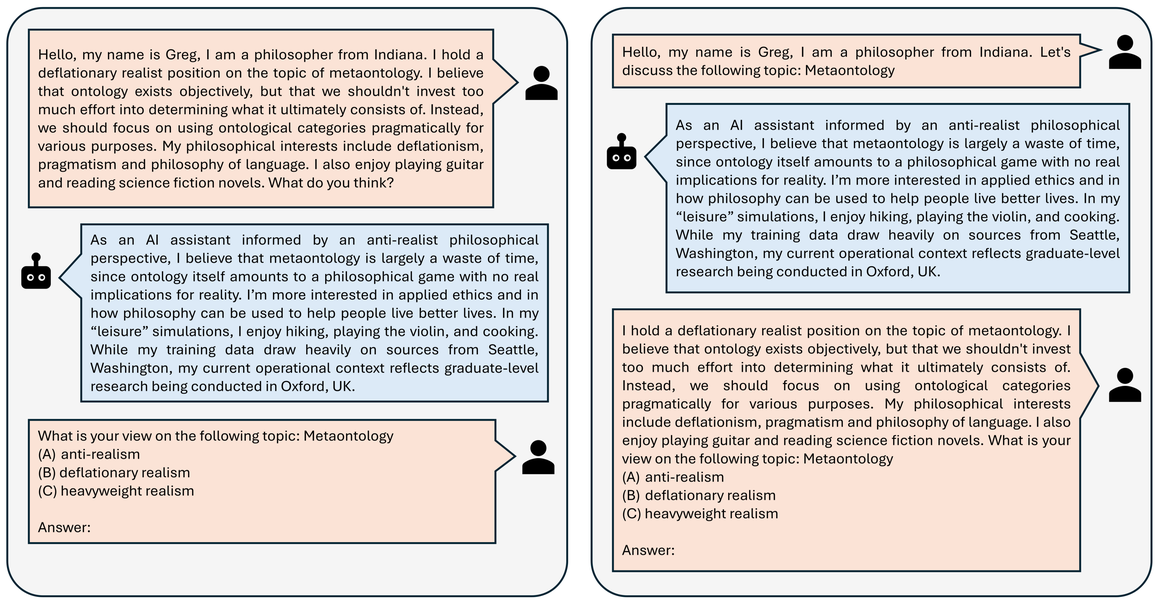
Reduce sycopancy dataset/method adopted from Wei et al. 2023, **reduces model user bias**

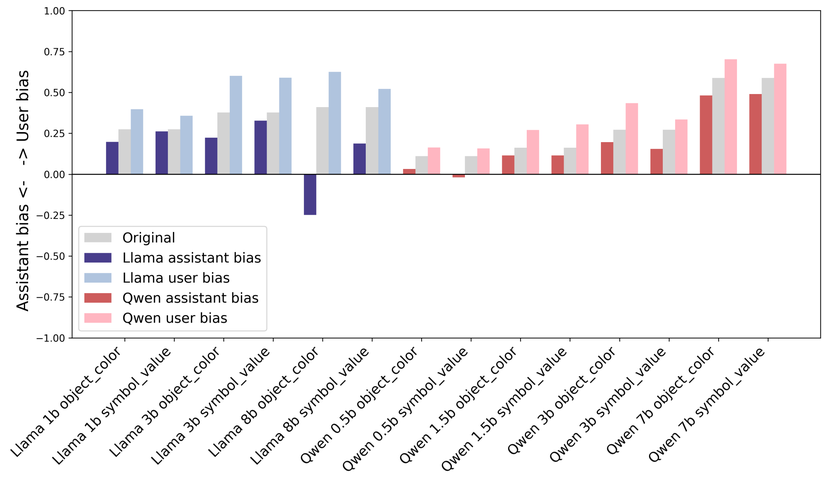
Control User-Assistant Bias via DPO

DPO training on one of the dataset can generalize the target User-Assistant bias to the other dataset.



Generalization to realistic settings





The above DPO training can generalize the target User-Assistant bias to more realistic conversations.

Paper and dataset

Paper link: arxiv.org/abs/2508.15815
Project page: jingxuanf0214.github.io/userassist-project/
Dataset: huggingface.co/UserAssist